INSTALLING AN OPERATING SYSTEM

Laboratory Report — Windows 11 Installation and Configuration in Oracle VirtualBox

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Lab environment: Oracle VirtualBox; Windows 11 Pro (64-bit)

1. Objective

To install Windows 11 in a virtual machine using Oracle VirtualBox, configure essential hardware and system settings, install Guest Additions, review updates and security, and verify system performance and successful operation.

2. Equipment and Software

Computer with VirtualBox; Windows 11 x64 ISO image; internet connection; Windows 11 virtual machine.

3. Virtual Machine Configuration

Operating system	Windows 11 (64-bit), Pro edition
Virtual memory	4096 MB (4 GB)
Virtual processors	2 CPU cores
Virtual disk	80 GB VDI
Firmware and security	EFI enabled; TPM 2.0; Secure Boot enabled
Network	NAT virtual adapter
Display memory	128 MB

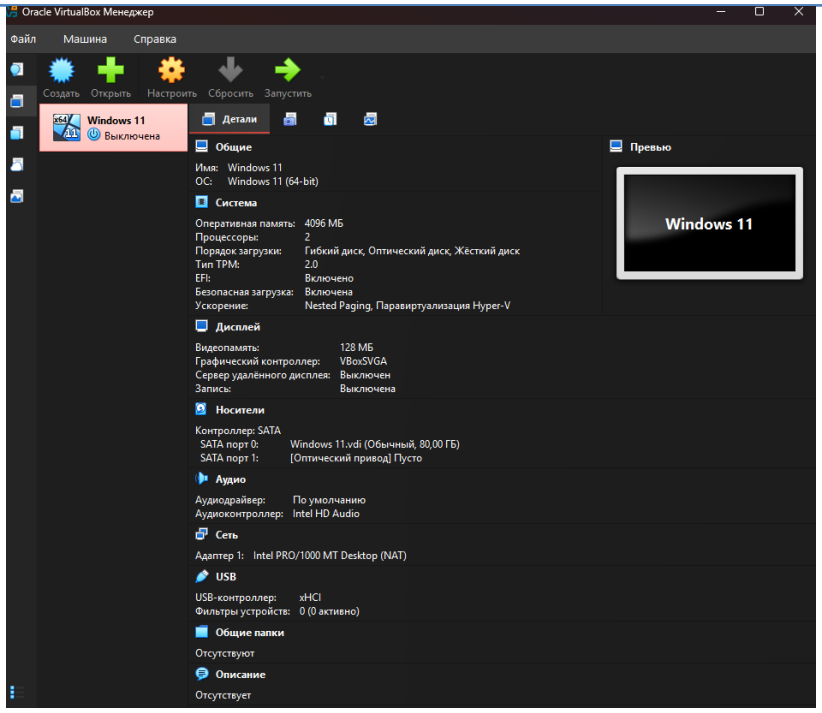


Figure 1. Virtual machine hardware configuration in VirtualBox.

4. Installation and Configuration Evidence

4.1 Booting from the installation ISO

The Windows 11 x64 ISO image was attached to the virtual optical drive. After the virtual machine started, the Windows setup language-selection screen appeared. An earlier ARM64 image was not suitable for this x64 virtual machine and was replaced.

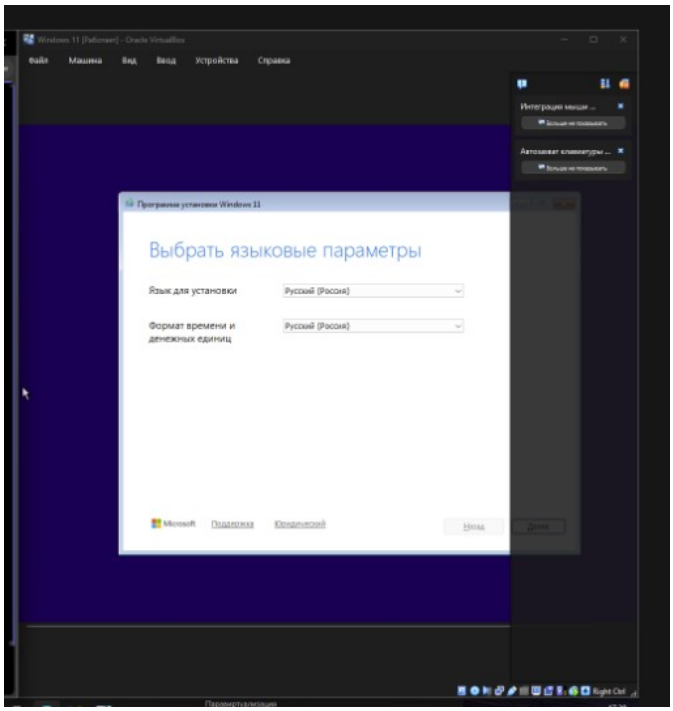


Figure 2. Windows 11 setup — language selection (boot/installation stage).

4.2 Installing Windows 11

Windows 11 Pro was selected. The setup copied and installed system files on the virtual disk, followed by automatic restarts and initial region/account configuration.

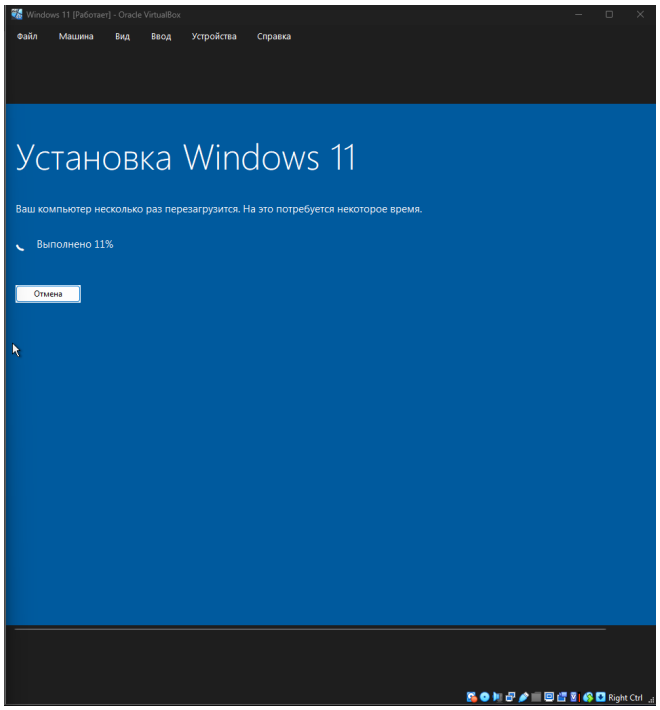


Figure 3. Windows 11 installation in progress (11%).

4.3 Installing VirtualBox Guest Additions

The Guest Additions virtual CD was mounted and the Windows installer was launched. The setup reported successful completion and requested a restart.

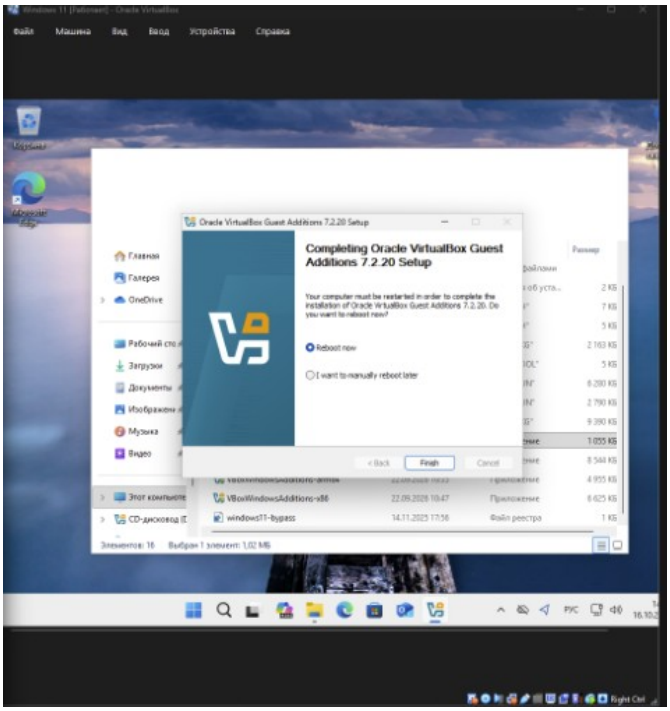


Figure 4. Oracle VirtualBox Guest Additions setup completed.

4.4 Windows Update

Windows Update was opened and available updates were identified. Installation and a restart were initiated. The screenshots show the update workflow; they do not independently prove that all updates finished successfully.

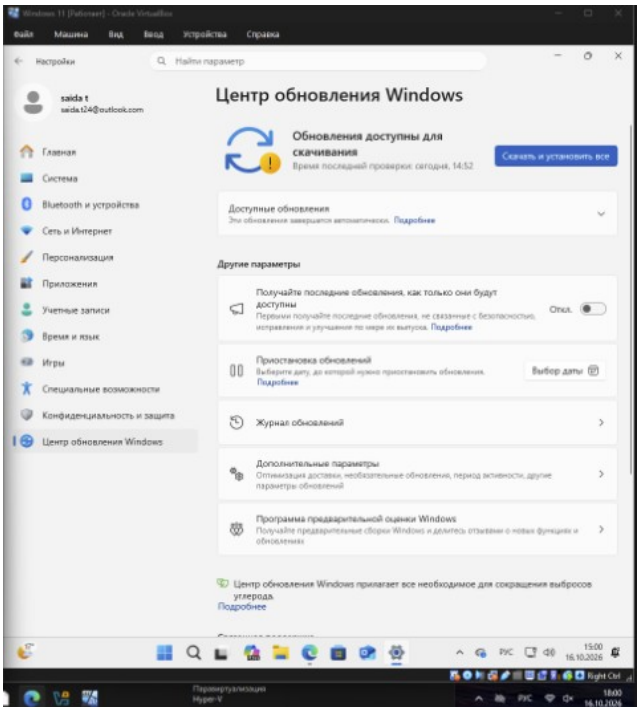


Figure 5. Windows Update displaying available updates.

4.5 Windows Security

Windows Security settings were reviewed. The security categories displayed normal green status indicators in the provided screenshot.

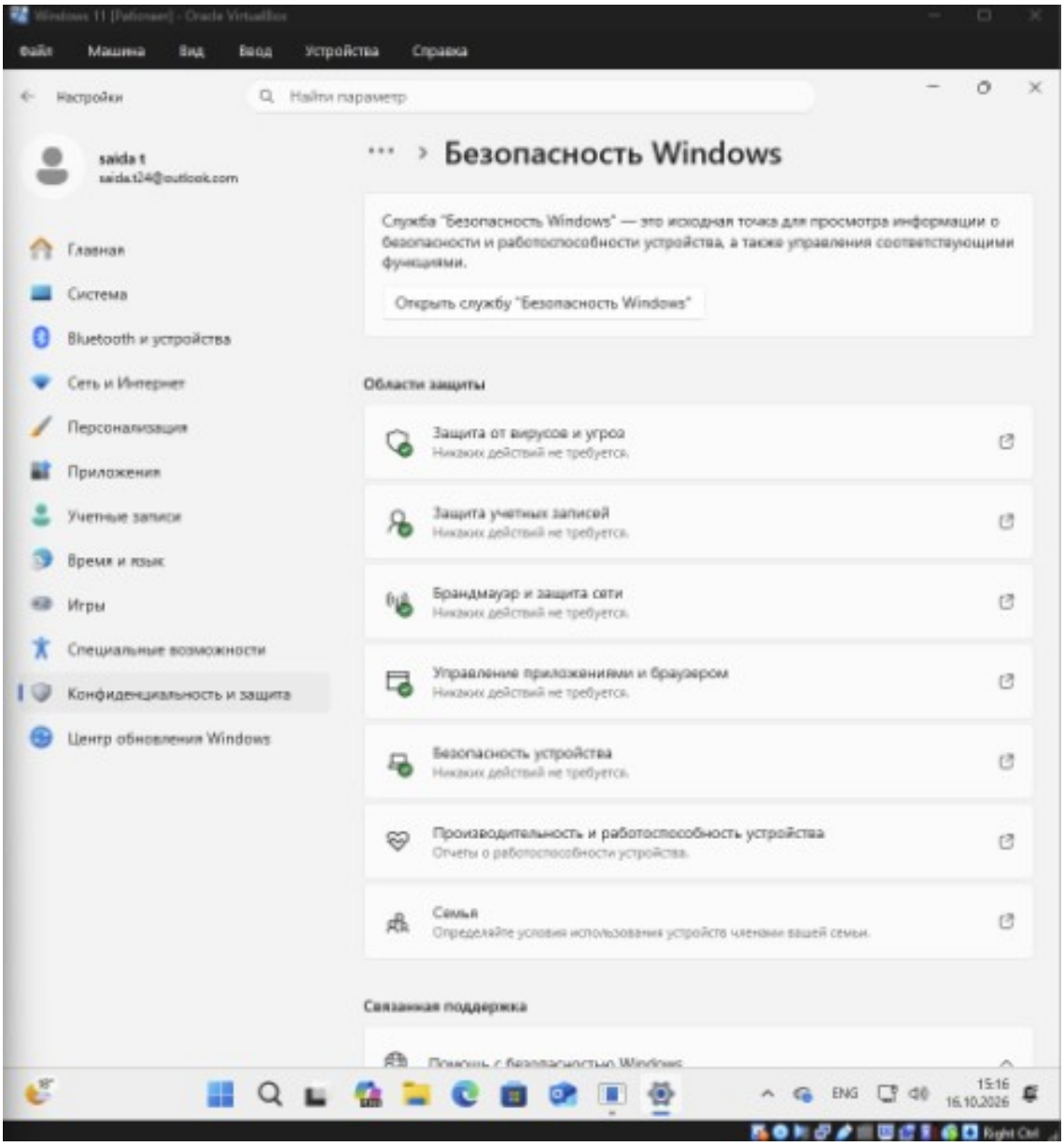


Figure 6. Windows Security settings and protection status.

4.6 Performance verification

Task Manager was opened to the Performance tab. CPU, memory, disk and network activity were visible. This confirms that performance monitoring was performed; a measured before/after optimization comparison was not provided.

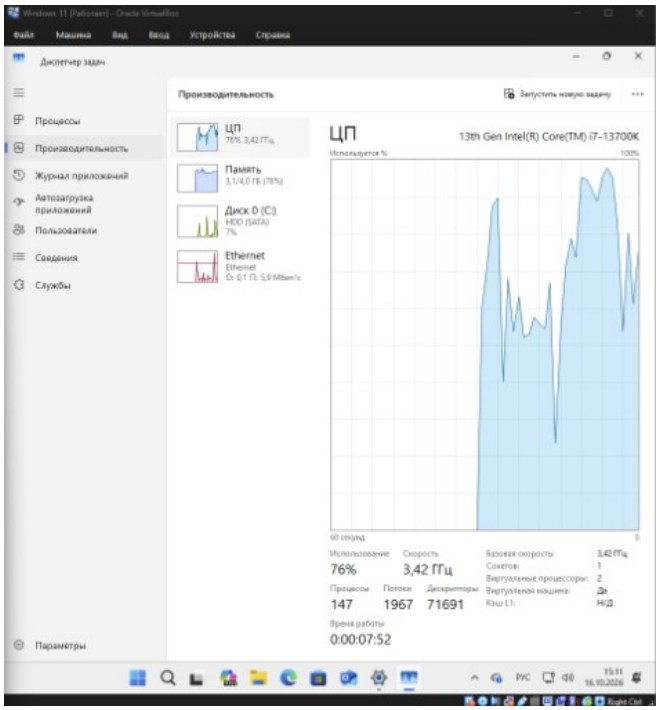


Figure 7. Task Manager — Performance / CPU.

4.7 Final working operating system

The virtual Windows 11 desktop was operational. The winver dialog confirmed Windows 11 Pro, version 26H2 (OS build 26300.9457), as shown in the captured environment.

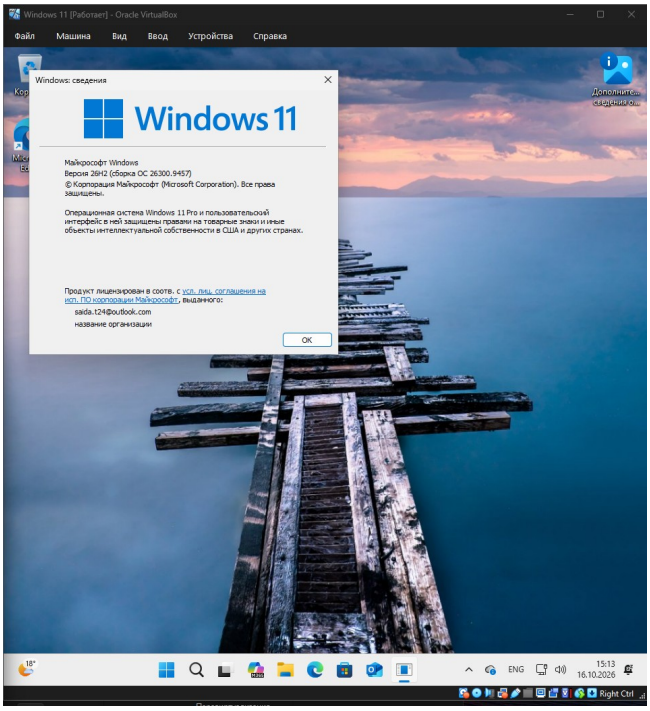


Figure 8. Final working Windows 11 Pro desktop and OS version.

5. Observations

• Virtual machine resources: 4 GB RAM, 2 CPU cores and an 80 GB virtual disk.

• Installation media: the ARM64 ISO was initially attached; a compatible x64 ISO was subsequently used.

• Installation progressed through language selection, edition selection, file installation and first-run setup.

• VirtualBox Guest Additions installation completed and prompted a restart.

• Windows Update detected available updates; a final fully-updated state was not documented.

• Windows Security showed green status indicators in the available capture.

• Task Manager displayed CPU and memory utilization; no before/after optimization measurements were recorded.

• Exact VM creation and OS installation durations were not recorded.

• No snapshot creation evidence was supplied; a snapshot should be created and documented if required by the instructor.

• Disk partition details and a full Device Manager driver check were not documented in the supplied screenshots.

6. Review Questions and Answers

1. Why is configuring the correct boot order in a virtual machine important?

It ensures the VM starts from the installation ISO during setup and from the installed virtual hard disk afterward.

2. What are the advantages of creating separate partitions during OS installation?

They can separate operating system files from user data and make organization, backup and recovery easier.

3. Why should chipset and system drivers be installed before other device-specific drivers?

They provide foundational hardware support on which other devices and drivers may depend.

4. How can you verify if all virtual hardware components are functioning properly?

Check Device Manager for warning icons, test network and display functionality, and inspect VirtualBox settings.

5. What security measures should be implemented after OS installation?

Install updates, keep built-in antivirus and firewall enabled, use secure account settings and create a restore point or VM snapshot.

6. How does the installation process differ between Windows and Linux?

Windows uses its setup wizard and Windows-specific configuration; Linux distributions use their own installers, package managers and partitioning options.

7. What is the purpose of using an ISO image in VirtualBox?

It acts as a virtual installation disc from which the operating system can boot and install.

8. What is the difference between a physical computer and a virtual machine?

A physical computer uses actual hardware, while a virtual machine uses virtualized hardware resources supplied by a host.

9. What are VirtualBox Guest Additions and what functions do they provide?

They are guest integration drivers and utilities that improve display handling, mouse integration, and features such as shared clipboard and folders.

10. Why is creating a virtual machine snapshot useful before performing major system changes?

A snapshot provides a recovery point so the VM can be returned to an earlier state if a change causes problems.

7. Conclusion

Windows 11 Pro was successfully installed and started in Oracle VirtualBox. The VM was configured with suitable virtual hardware, Guest Additions was installed, and Windows Update, security settings and system performance were inspected. The screenshots confirm the main installation stages and a functioning final operating system. A fully completed update status, snapshot and quantitative optimization comparison would require additional evidence if specifically requested.